

Safety



TIMESTAMP

Scalar Functions: Date/Time

[PARSE_TIMESTAMP](#)

[FORMAT_TIMESTAMP](#)

Raw Streams

```
-----  
-- STREAM raw_failures  
-- topic generado por simulador  
-----  
  
DROP STREAM IF EXISTS `raw_failures`;  
CREATE STREAM `raw_failures` (  
  `id` VARCHAR,  
  `equipment` VARCHAR,  
  `station` VARCHAR,  
  `location` VARCHAR,  
  `fts` VARCHAR,  
  `duration` DOUBLE  
)  
WITH (  
  KAFKA_TOPIC='failures',  
  VALUE_FORMAT='JSON');  
  
DESCRIBE `raw_failures` EXTENDED;  
SELECT * FROM `raw_failures`;  
  
-----  
-- STREAM raw_incidents  
-- topic generado por simulador  
-----  
  
DROP STREAM IF EXISTS `raw_incidents`;  
CREATE STREAM `raw_incidents` (  
  `id` VARCHAR,  
  `type` VARCHAR,  
  `station` VARCHAR,  
  `location` VARCHAR,  
  `its` VARCHAR,  
  `severity` VARCHAR,  
  `duration` DOUBLE  
)
```

```

WITH (
  KAFKA_TOPIC='incidents',
  VALUE_FORMAT='JSON');

DESCRIBE `raw_incidents` EXTENDED;
SELECT * FROM `raw_incidents`;

```

Staging failures: stream stg_failures, table failures

```

--
-- Stream particionado monitor de fallos
DROP STREAM IF EXISTS `stg_failures`;
CREATE STREAM `stg_failures`
WITH (kafka_topic='stg_failures', key_format='AVRO', value_format='AVRO')
AS SELECT STRUCT(`equipment` := `equipment`, `station` := `station`) as fk,
           `location`,
           PARSE_TIMESTAMP(`fts`, 'yyyy-MM-dd'T'HH:mm:ss') as `fts`,
           `duration`
FROM `raw_failures`
-- Filtrado de nulos
WHERE `equipment` IS NOT NULL AND `station` IS NOT NULL AND
      `location` IS NOT NULL AND `ts` IS NOT NULL AND `duration` IS NOT NULL
PARTITION BY STRUCT(`equipment` := `equipment`, `station` := `station`);

print `stg_failures` from beginning;
DESCRIBE `stg_failures` EXTENDED;
SELECT * FROM `stg_failures`;

--
-- Tabla fallos totales (estación, equipo, hora)
CREATE TABLE `failures`
WITH (kafka_topic='stfail', KEY_FORMAT='AVRO', VALUE_FORMAT='AVRO')
AS SELECT fk->`station` as `station`,
           fk->`equipment` as `equipment`,
           FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid') as `hh`,
           count(*) as `nf`, sum(`duration`) as `df`
FROM `stg_failures`
GROUP BY fk->`station`, fk->`equipment`, FORMAT_TIMESTAMP(`fts`, 'HH',
'Europe/Madrid');

print `stfail` from beginning;
DESCRIBE `failures` EXTENDED;
SELECT * FROM `failures`;

-- Tabla fallos totales (estación, hora)
CREATE TABLE `fastho`
WITH (kafka_topic='fastho', KEY_FORMAT='AVRO', VALUE_FORMAT='AVRO')
AS SELECT fk->`station` as `station`,
           FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid') as `hh`,
           count(*) as `nf`, sum(`duration`) as `df`
FROM `stg_failures`
-- Sin WHERE
GROUP BY fk->`station`, FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid');

```

Staging incidents: stream stg_incidents, table incidents

```
--  
-- Stream particionado monitor de incidentes  
DROP STREAM IF EXISTS `stg_incidents`;  
CREATE STREAM `stg_incidents`  
WITH (kafka_topic='stg_incidents', key_format='AVRO', value_format='AVRO')  
AS SELECT STRUCT(`type` := `type`, `station` := `station`) as ik,  
           `location`,  
           PARSE_TIMESTAMP(`its`, 'yyyy-MM-dd' 'T' 'HH:mm:ss') as `its`,  
           `severity`,  
           `duration`  
FROM `raw_incidents`  
-- Filtrado de nulos  
WHERE `type` IS NOT NULL AND `station` IS NOT NULL AND  
      `location` IS NOT NULL AND `ts` IS NOT NULL AND  
      `severity` IS NOT NULL AND `duration` IS NOT NULL  
PARTITION BY STRUCT(`type` := `type`, `station` := `station`);  
  
print `stg_incidents` from beginning;  
DESCRIBE `stg_incidents` EXTENDED;  
SELECT * FROM `stg_incidents`;  
  
--  
-- Tabla incidentes totales (estación, tipo, hora)  
CREATE TABLE `incidents`  
WITH (kafka_topic='stinci', KEY_FORMAT='AVRO', VALUE_FORMAT='AVRO')  
AS SELECT fk->`station` as `station`,  
          fk->`type` as `type`,  
          FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid') as `hh`,  
          count(*) as `nf`, sum(`duration`) as `df`  
FROM `stg_incidents`  
-- Sin WHERE  
GROUP BY fk->`station`, fk->`type`, FORMAT_TIMESTAMP(`fts`, 'HH',  
'Europe/Madrid');  
  
-- Tabla fallos totales (estación, hora)  
CREATE TABLE `fastho`  
WITH (kafka_topic='fastho', KEY_FORMAT='AVRO', VALUE_FORMAT='AVRO')  
AS SELECT fk->`station` as `station`,  
          FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid') as `hh`,  
          count(*) as `nf`, sum(`duration`) as `df`  
FROM `stg_incidents`  
GROUP BY fk->`station`, FORMAT_TIMESTAMP(`fts`, 'HH', 'Europe/Madrid');
```